Technology

# Broken gadget? Simply dissolve it

3D-printed devices that break up in water could make electronics more sustainable

Jeremy Hsu

ELECTRONICS such as Bluetooth speakers can be 3D-printed from a material that dissolves in water within hours. That allows designers to rapidly create prototypes, enables easier recycling of the resulting electronic waste, and may even inspire more sustainable versions of mass-manufactured consumer electronics.

Researchers demonstrated the dissolvable technology in printed circuit boards, which contain the crucial components and wiring of modern electronics. Hundreds of millions of printed circuit boards are manufactured every year for military fighter jets, cars, medical devices, smartphones and cheap toys. But the world recycles just a fraction of these devices in a "very brute-force way", by shredding them to extract reusable materials, says Huaishu Peng at the University of Maryland.

Peng and his colleagues designed 3D-printable circuit boards using polyvinyl alcohol, a polymer that can dissolve in water. To form the wiring, they injected a gallium-indium metal alloy into the circuit board's channels in liquid form. They then manually placed electronic components on the board. Additional polymer glue was applied to seal the circuits before drying the device for an hour at 60°C.

**"Dissolvable circuit boards could be very useful when designers are testing electronics prototypes"**

Using these boards, the researchers assembled working versions of a Bluetooth speaker, a fidget toy and an electronic three-finger gripper. A little splash of water won't instantaneously destroy such devices – but after 36 hours in room-temperature 22°C water, the machines did dissolve.

Then the researchers easily picked out the electronic components and most of the liquid metal, which had broken up into small beads. Once they evaporated the water, they were also able to recover 99 per cent of the dissolved polyvinyl alcohol (*ACM Symposium on User Interface Software and Technology*, DOI: 10.1145/3746059.3747604).

Such dissolvable circuit boards could be very useful when designers are rapidly building and testing electronics prototypes, because it is usually difficult to recycle printed circuit boards, says Jasmine Lu at the University of Chicago in Illinois, who has done related research on reusing circuit board materials.

A 2022 United Nations report showed Asia generated 600,000 tonnes of used circuit boards while recycling just 17 per cent of this type of e-waste. Europe and North America generated 300,000 tonnes of printed circuit boards apiece, with Europe managing to recycle 61 per cent of this e-waste and North America 44 per cent.

The fact anyone with a 3D printer can adopt this dissolvable electronics approach makes it especially unique when compared to other sustainable electronics efforts, says Lu. During use, the devices could be further protected with temporary waterproof cases, Peng suggests.

But the limited durability of the circuit boards currently makes the dissolvable electronics better suited for the rapid prototyping of designs, rather than the mass manufacture of finished electronics products, says Lu.

Peng and his colleagues are contacting circuit board producers to explore how mass manufacturing might work. ▮

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Palaeontology

## Dinosaur's bizarre spikes were fused to its skeleton

**A DINOSAUR fossil found in Morocco may be the most bizarrely and elaborately armoured vertebrate that has ever walked the planet.**

**The first fossil of *Spicomellus afer* was discovered in Morocco and reported in 2021. It was only a rib fragment with fused spikes, suggesting that it belonged to a group of dinosaurs known as ankylosaurs. These short-limbed, wide-bodied herbivorous dinosaurs are characterised by their covering of plates and spines.**

**Then, in October 2022, a farmer in the badlands of Morocco's Middle Atlas mountains began to excavate a much more complete *Spicomellus* skeleton. That fossil has now been dated to 165 million years ago, in the Jurassic Period. The creature was probably about 4 metres long and weighed up to 2 tonnes.**

THE TRUSTEES OF THE NATURAL HISTORY MUSEUM, LONDON

A *Spicomellus afer* fossil which was uncovered during a dig in Morocco

**Armoured dinosaurs such as stegosaurs and ankylosaurs, like modern crocodiles, had bony plates that sat in the skin called osteoderms. But in the *Spicomellus* fossil, there are two different types of bony armour: osteoderms and spikes that are actually fused to the bone (*Nature*, doi.org/p35j).**

**"It's unheard of among armoured dinosaurs, and indeed anything that has osteoderms, which is totally crazy," says Susannah Maidment at the Natural History Museum in London, a member of the team that analysed the fossil. The creature was so bizarre, Maidment says she "ran out of hyperboles to describe it."**

**In total, the *Spicomellus* specimen has dozens of armoured spikes covering almost its entire body. Some spikes attached to a neck collar are nearly a metre in length. There are also fused vertebrae in the tail, indicating that it may have been a fierce weapon.**

**The armour is so complex that the researchers think it may also have been used to attract mates. ▮**

James Woodford